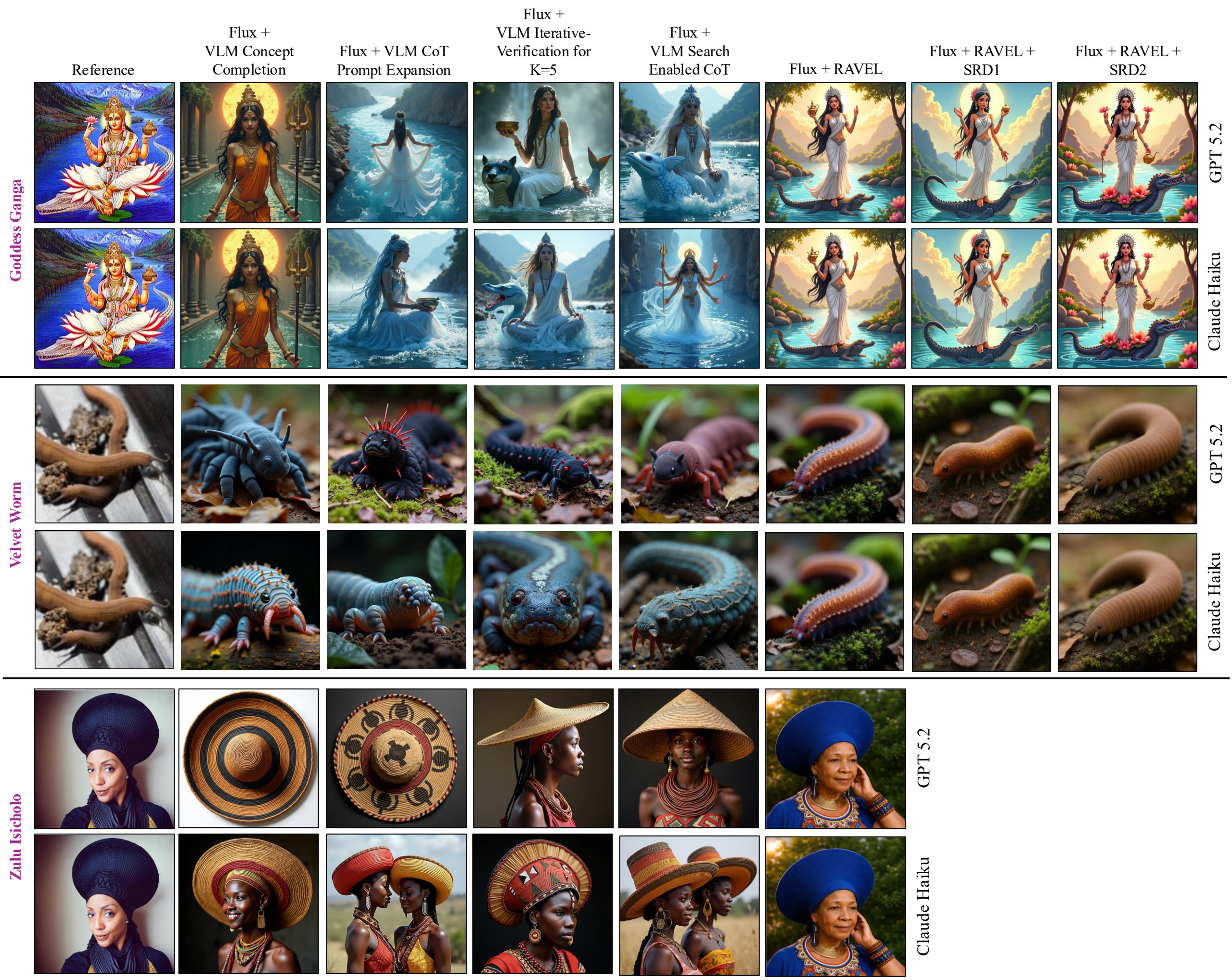

Figure R3. Qualitative comparison of frontier LLM pipelines vs. RAVEL (Flux backbone). Three rare concepts spanning mythology (Goddess Ganga), biology (Velvet Worm), and cultural artifacts (Zulu Isicholo), evaluated across GPT-5.2 and Claude Haiku pipelines with increasing augmentation. Concept Completion collapses to generic priors, Ganga rendered as a plain woman, Velvet Worm as a blue armadillo-like creature, Isicholo as a decorative hat without cultural context. CoT Prompt Expansion and Iterative Verification (K=5) improve partially but fail to recover defining attributes: Ganga's multi-armed form and makara vehicle, the Velvet Worm's characteristic velvety texture and claw-tipped legs, and the Isicholo's distinctive flat-topped silhouette and woven grass construction. Search-enabled CoT shows further gains but remains inconsistent across model families. In contrast, Flux + RAVEL recovers concept-specific attributes in a single pass; for Zulu Isicholo, the GSI threshold is reached at the initial RAVEL output, demonstrating early convergence without requiring SRD refinement. For the remaining concepts, SRD rounds provide targeted iterative refinement. These results confirm that the performance gap in Table R1 reflects a structural limitation of ungrounded LLM pipelines rather than model capability.

**Reference images are provided for comparison purposes only and are not used during generation.*